

M. R. Tanush Pai

AI/ML Engineer · Python Developer · Generative AI & LLM Systems · Agentic AI · Backend & REST APIs
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PROFESSIONAL SUMMARY

AI/ML Engineer (B.Tech, Computer Science – Data Science) with professional experience building **Generative AI**, **LLM**, and **agentic systems** for enterprise applications. Experienced in developing end-to-end AI solutions involving **LLMs**, natural-language data querying, document intelligence, and AI-powered analytics, with strong **Python**, **SQL**, and **backend engineering** skills.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, R

AI & GenAI: LLMs, RAG, Agentic AI, LangChain, LangGraph, Prompt Engineering, Hugging Face, OpenAI SDK

Data & Databases: Pandas, NumPy, PostgreSQL, MongoDB, Redis

Backend & APIs: FastAPI, REST APIs, Prisma

Cloud & DevOps: AWS, Docker, Google Cloud (Vertex AI), Vercel

Tools: Git, GitHub, Bitbucket, Jira

PROFESSIONAL EXPERIENCE

Jr. AI Engineer | Sam Corporate

Kochi, India

Aug 2026 – Present

- Designed and built an end-to-end PoC for an enterprise Nextcloud **RAG** platform, orchestrating LangGraph **agents** with **zero PII leakage**.
- Integrated **Apache Superset** into the ESG Reporting application as an embedded **BI** solution, replacing Power BI for dashboarding and reducing significant licensing costs; implemented dynamic **RLS** based on user IDs with workspace-level data isolation.
- Designed and implemented a scalable **batch document-processing microservice** using **Gemini on Google Cloud Vertex AI** to semantically extract and normalize information from bills, mapping varying field labels to standardized schema fields and returning **structured JSON** while concurrently processing **100+ PDFs**.

AI/ML Engineer Intern | Sam Corporate

Kochi, India

Feb 2026 – July 2026

- Architected an **agentic AI** analytics chatbot with an **LLM-based planner** that dynamically generates **MongoDB** aggregation pipelines across **25+ collections**, enabling non-technical users to query data using natural language with GDPR-aligned privacy controls.
- Built an AI-powered **QMS document generation platform** adopted by **12+ employees**, automating content generation across **10+ templates** using LLMs and reducing document turnaround from hours to minutes.
- Extended an **ESG analytics platform** with **8+ modules**, including AI-driven **report generation**, **NL2SQL**, peer benchmarking, and interactive **data visualization**, using **PostgreSQL**, OpenAI API, and Perplexity API.
- Improved **production LLM reliability** by optimizing **prompt engineering** and API workflows, eliminating **HTTP 429** rate-limit errors and stabilizing application performance.
- Developed an **automated MongoDB data validation tool** that cross-checks records across multiple databases and generates **Excel reports**, replacing manual validation workflows with a single-click process.

Data Analytics Trainee | Navodita Infotech

Remote

Aug 2025 – Sep 2025

- Built a **collaborative filtering recommendation system** processing **10K+ user-movie ratings** using **Python**, **Pandas**, cosine similarity, and **NumPy**, generating Top-10 personalized recommendations per user.

Data Science Trainee | Keltron REC

Kochi, India

Jun 2025

- Preprocessed **5+ real-world datasets**, built **Power BI** dashboards, and applied **ML fundamentals** using Excel and **SQL**.

KEY PROJECTS

AVANA – Autonomous Visual Analytics & Normative Agents | *Python, LangGraph, LangChain, Ollama, FLAML*

- Architected a **3-agent LangGraph pipeline** (*Data Engineer* → *Data Analyst* → *Data Scientist*) that autonomously transforms raw CSV uploads into cleaned datasets, **EDA** reports, and trained **ML models** with zero manual intervention.
- Implemented a **Master Orchestrator** with a shared Dataset Contract (target column, problem type, domain context) coordinating agents across a **13-stage preprocessing pipeline**, **50+ automated visualizations**, and **FLAML AutoML** training.
- Integrated an **AI Consultant** chat interface (Ollama) with role-aware personas for contextual interaction with pipeline outputs, visualizations, and model insights.

EDUCATION

SCMS School of Engineering and Technology

B.Tech in Computer Science (Data Science)

- Affiliated to APJ Abdul Kalam Technological University
- CGPA: 7.92

Kerala, India

Oct 2022 – Apr 2026

CERTIFICATIONS

- Multi AI Agent Systems with CrewAI – DeepLearning.AI, Coursera (2025)
- Fundamentals of Building AI Agents – IBM, Coursera (2025)
- Data Analysis with Python and Pandas – Infosys Springboard (2025)
- Python for Data Science – Infosys Springboard (2025)